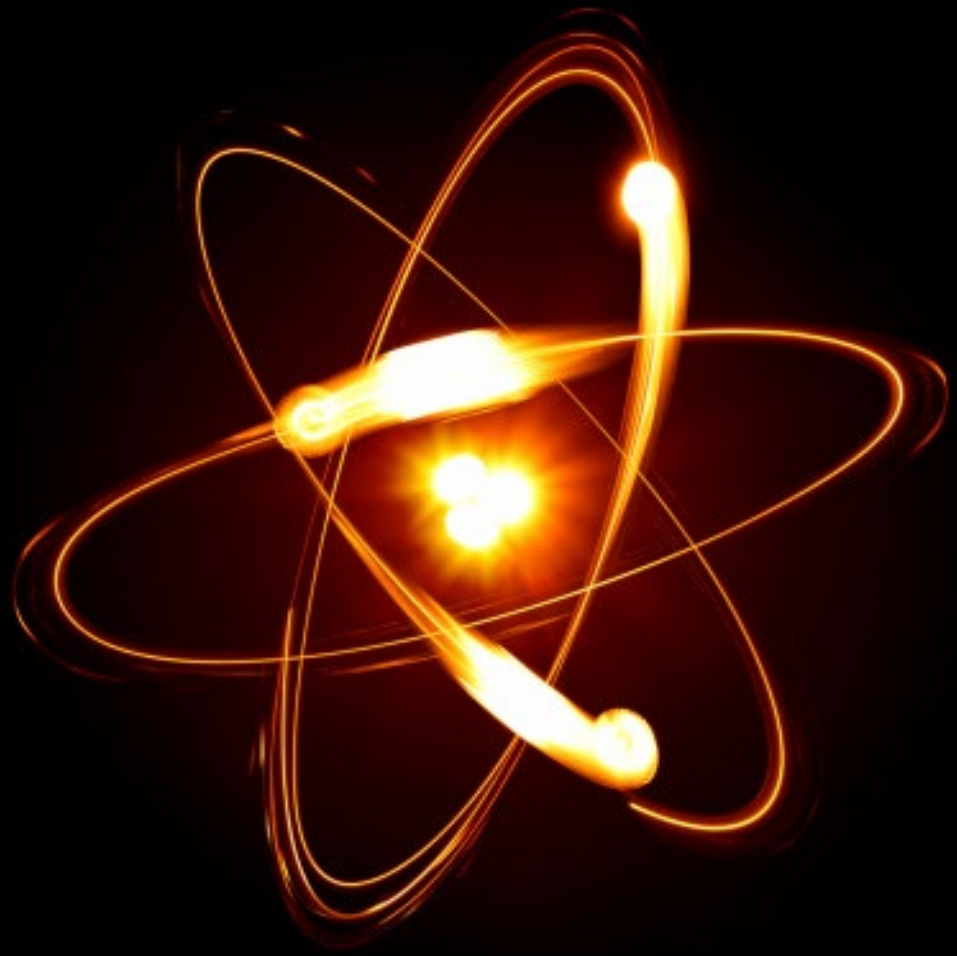
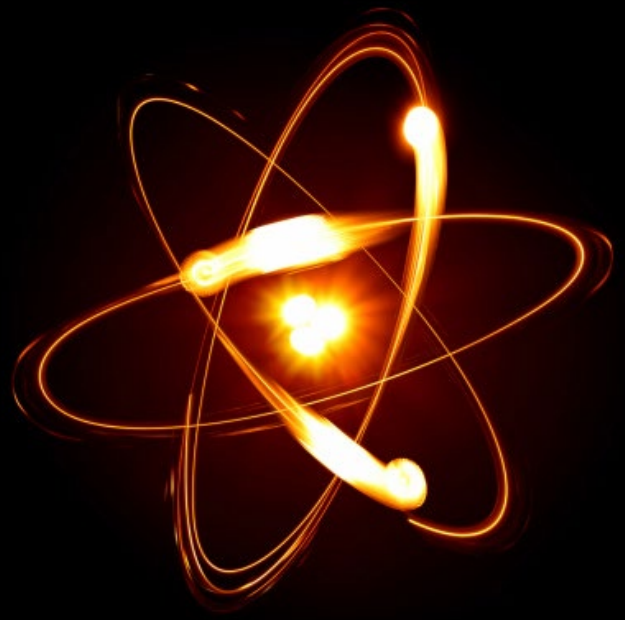
Chapter 19.

General angular momentum



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Quantum Physics

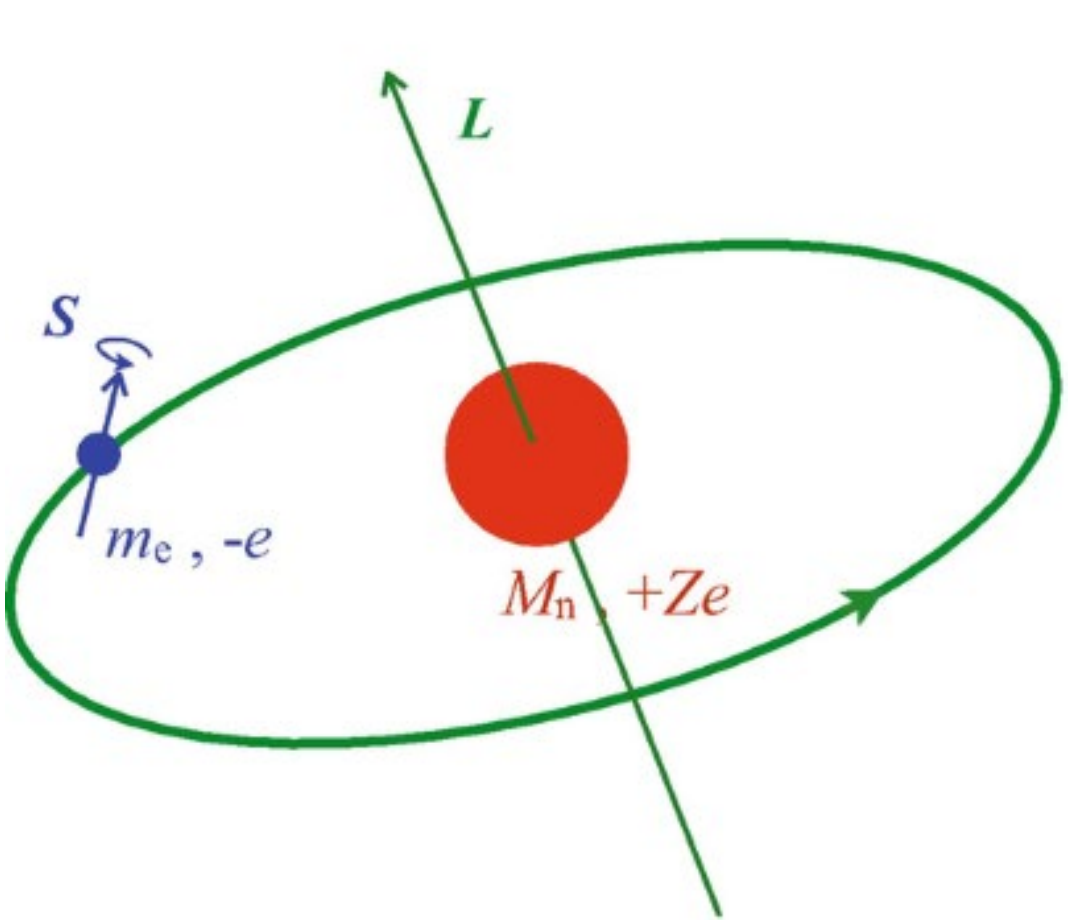


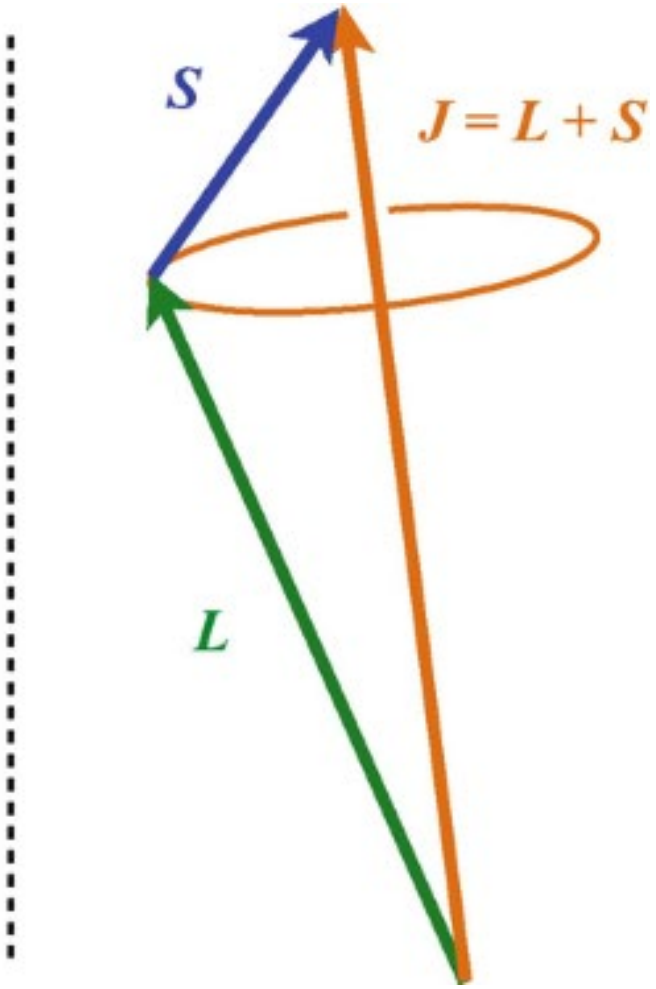
Chapter 19.

General angular momentum

- Definition and notation
- Eigenvalues of J^2 and J_z. Lemmas
- Spectrum of J^2 and J_z
- $\{|k,j,m\rangle\}$ representation
- The spaces $\mathcal{E}(k,j)$
- Standard basis

General angular momentum.





General angular momentum[2]

General angular momentum

Let us consider a vector operator $\vec{J} = \vec{\imath}J_x + \vec{\jmath}J_y + \vec{k}J_z$, acting on some state space.

The angular momentum is defined as any set of three observables, $J_x,\ J_y,\ J_z$ which satisfy the following commutation relations

$$[J_x, J_y] = i\hbar J_z \qquad [J_y, J_z] = i\hbar J_x \qquad [J_z, J_x] = i\hbar J_y$$

Now, $J^2 = J_x^2 + J_y^2 + J_z^2$ is hermitian and we assume it is an observable. It can be shown that

$$[J^2, J_x] = [J^2, J_y] = [J^2, J_z] = 0$$

What are the common eigenvectors of J^2 and J_z and the corresponding eigenvalues?

Obviously the "orbital" angular momentum (previously studied) is such a kind of operator, but are there other possible operators which behave like this?

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General angular momentum

Main purpose: Determine the spectrum of J^2 and J_z and their common eigenvectors

Let us define: $J_+ = J_x + iJ_y$ $J_- = J_x - iJ_y$

Obviously $J_+^\dagger = J_-$ and $J_-^\dagger = J_+$

It is easy to show that:

- $[J_z, J_+] = \hbar J_+$:

$$[J_z, J_+] = J_z(J_x + iJ_y) - (J_x + iJ_y)J_z = J_zJ_x + iJ_zJ_y - J_xJ_z - iJ_yJ_z = [J_z, J_x] + i[J_z, J_y] = i\hbar J_y + i(-i\hbar J_x) = \hbar(J_x + iJ_y) = \hbar J_+$$

- $[J_z, J_-] = -\hbar J_-$
- $[J_+, J_-] = 2\hbar J_z$
- $[J^2, J_+] = [J^2, J_-] = [J^2, J_z] = 0$

- $J_+J_- = J^2 - J_z^2 + \hbar J_z$
- $J_-J_+ = J^2 - J_z^2 - \hbar J_z$
- $J^2 = \frac{1}{2}(J_+J_- + J_-J_+) + J_z^2$

Eigenvalues of J^2 and J_z

All the eigenvalues of J^2 are positive or zero

Consider an arbitrary state $|\Psi\rangle \in \mathcal{E}$. The expectation value of J^2 is $\langle\Psi|J^2|\Psi\rangle \geq 0$

If $|\Psi\rangle$ is an eigenstate of J^2, the expectation value is the eigenvalue multiplied by the square of the norm of $|\Psi\rangle$, and since this is non negative we can write it uniquely as $\hbar^2 j(j+1)$ with j a non negative real value.

As for the eigenvalues of J_z , we write them as $m\hbar$

Since $[J^2, J_z] = 0$ we can find a set of eigenstates common to J^2 and J_z . Let this set be $\{|k, j, m\rangle\}$, where j specifies the eigenvalue of J^2, namely $\hbar^2 j(j+1)$, m specifies the eigenvalue of J_z, namely $m\hbar$, where m is a real number, and k denotes the different eigenstates with the same values of j and m that can appear due to the possible degeneracy.

$$J^2|k, j, m\rangle = \hbar^2 j(j+1)|k, j, m\rangle \qquad J_z|k, j, m\rangle = m\hbar|k, j, m\rangle$$

Eigenvalues of J^2 and J_z. Lemma I

If $j(j+1)\,\hbar^2$ and $m\hbar$ are the eigenvalues of J^2 and J_z associated to the same eigenvector $|k,j,m\rangle$, then $-j \le m \le j$

Let $|\Phi\rangle = J_-|k,j,m\rangle$. Then (we assume $|k,j,m\rangle$ is normalized)

$\langle\Phi| = \langle k,j,m|J_-^\dagger = \langle k,j,m|J_+$. Therefore

$$\||J_-|k,j,m\rangle\|^2 = \langle k,j,m|J_+J_-|k,j,m\rangle = \langle\Phi|\Phi\rangle \ge 0$$

$$0 \le \langle k,j,m|J_+J_-|k,j,m\rangle = \langle k,j,m|J^2 - J_z^2 + \hbar J_z|k,j,m\rangle = \hbar^2[j(j+1) - m^2 + m] = \hbar^2(j-m+1)(j+m) \qquad (*)$$

and as a consequence: $-j \le m \le j+1$

$$J_+J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 - iJ_xJ_y + iJ_yJ_x = J^2 - J_z^2 - i[J_x, J_y] = J^2 - J_z^2 + \hbar J_z$$

Eigenvalues of J^2 and J_z. Lemma I

We now consider | Φ >= J_+ | k, j, m >. Then

- Similarly

$$0 \leq \langle k,j,m|J_-J_+|k,j,m\rangle = \langle k,j,m|J^2 - J_z^2 - \hbar J_z|k,j,m\rangle = \hbar^2[j(j+1) - m^2 - m] = \hbar^2(j+m+1)(j-m) \qquad (**)$$

$$-(j+1) \leq m \leq j$$

Combining both inequalities we get

$$-j \leq m \leq j$$

$$J_-J_+ = (J_x - iJ_y)(J_x + iJ_y) = J_x^2 + J_y^2 + iJ_xJ_y - iJ_yJ_x = J^2 - J_z^2 + i[J_x, J_y] = J^2 - J_z^2 - \hbar J_z$$

Eigenvalues of J^2 and J_z. Lemma II

Properties of the vector $J_-|k,j,m\rangle$

Let $|k,j,m\rangle$ be an eigenvector of J^2 and J_z with the eigenvalues $j(j+1)\hbar^2$ and $m\hbar$. Then

i) If $m=-j$, then $J_-|k,j,m\rangle=0$

ii) If $m>j$, then $J_-|k,j,m\rangle$ is a non-null eigenvector of J^2 and J_z with the eigenvalues $j(j+1)\hbar^2$ and $(m-1)\hbar$

- $J_-|k,j,-j\rangle=0$

  Taking into account $(*)$ for $m=-j$,

$$\langle k,j,-j|J_+J_-|k,j,-j\rangle=\hbar^2[j(j+1)-j^2-j]=0$$

Eigenvalues of J^2 and J_z. Lemma II

ii) If $m > j$, then $J_-|k,j,m\rangle$ is a non-null eigenvector of J^2 and J_z with the eigenvalues $j(j+1)\hbar^2$ and $(m-1)\hbar$

- If $-j < m$ then $J_-|k,j,m\rangle$ is an eigenvector of J^2 with eigenvalue $j(j+1)\hbar^2$

$$J^2 J_-|k,j,m\rangle = J_- J^2|k,j,m\rangle = J_-\big(\hbar^2 j(j+1)\big)|k,j,m\rangle = \hbar^2 j(j+1) J_-|k,j,m\rangle$$

- $J_-|k,j,m\rangle$ is an eigenvector of J_z with eigenvalue $(m-1)\hbar$.

$$J_z J_-|k,j,m\rangle = (-\hbar J_- + J_- J_z)|k,j,m\rangle = -\hbar J_-|k,j,m\rangle + \hbar m J_-|k,j,m\rangle = \hbar(m-1) J_-|k,j,m\rangle$$

$[J_z, J_-] = -\hbar J_-$ $\qquad$ $J_z J_- = -\hbar J_- + J_- J_z$

Eigenvalues of J^2 and J_z. Lemma III

Properties of the vector $J_+|k,j,m\rangle$

Let $|k,j,m\rangle$ be an eigenvector of J^2 and J_z with the eigenvalues $j(j+1)\hbar^2$ and $m\hbar$. Then

i. If $m = j$, then $J_+|k,j,j\rangle = 0$

ii. If $m < j$, then $J_+|k,j,m\rangle$ is a non-null eigenvector of J^2 and J_z with the eigenvalues $j(j+1)\hbar^2$ and $(m+1)\hbar$

Summary:

- $J_\pm|k,j,\pm j\rangle = 0$
- $J_\pm|k,j,m\rangle$ is an eigenvector of J^2 with eigenvalue $\hbar^2 j(j+1)$
- $J_\pm|k,j,m\rangle$ is an eigenvector of J_z with eigenvalue $\hbar(m \pm 1)$

Spectrum of J^2 and J_z

Consider any state of the form $|k, j, m\rangle$, with $-j \leq m \leq j$

Repeated application of J_- leads to eigenvectors of J^2 with eigenvalue $\hbar^2 j(j+1)$ and of J_z with eigenvalues $\hbar(m-1), \hbar(m-2), \ldots, \hbar(m-p)$.

Let *p* be the integer such that $-j \leq m - p < -j + 1$.

Now there are two possibilities:

1) If $m - p = -j$ then further applications of J_- will give zero.

2) On the other hand if the equality is not fulfilled, i.e., $-j < m - p < -j + 1$, then one further application of J_- will produce an eigenvector of J_z with eigenvalue $\hbar(m - p - 1)$;

   but $m - p - 1 < -j$, **which is impossible.**

Therefore we must have the relation

$$m - p = -j \qquad \text{where } p \text{ is an integer.}$$

Spectrum of J^2 and J_z

If instead of applying J_- we apply J_+ repeatedly, following the same arguments we conclude that

$$m + q = j$$

where q is an integer.

So we have

$$m + q = j \qquad m - p = -j \qquad \Rightarrow \qquad p + q = 2j$$

and therefore,

$$j = \frac{p+q}{2} = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$$

Also, the values of m for each j are,

$$m = -j, -j+1, \ldots, j-1, j$$

General angular momentum. Summary

- The square of a general angular momentum is quantized, it can only take values of the form

$$\hbar^2 j(j+1)$$

with $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, \ldots$

- The z component, J_z , of a general angular momentum is also quantized, with values of the form $\hbar m$ being $-j \leq m \leq j$.

- The orbital angular momentum $\vec{L}$ is a particular case of a general angular momentum, for which only integer values are allowed for the quantum number associated to $L^2(l)$

General angular momentum. $\{|k,j,m\rangle\}$ representation

Objective.- To construct an orthonormal basis in $\mathcal{E}_r$ with eigenvectors common to J^2 and J_z .

Take the eigenvalues $j(j+1)\hbar^2$ and $m\hbar$. The set of eigenvectors associated to these eigenvalues form a subspace $\mathcal{E}(j,m)$.

Its dimension is $g(j,m)$ which may be greater than one because J^2 and J_z are not a CSCO.

Assume we have chosen $\{|k,j,m\rangle, k = 1,2,\ldots,g(j,m)\}$ as an orthonormal basis in $\mathcal{E}(j,m)$.

For instance in the hydrogen atom a basis of $\mathcal{E}(0,0)$ is given by the orbitals $1s, 2s, 3s, \cdots$; a basis of $\mathcal{E}(1,-1)$ is given by the states with $n = 2, 3, \cdots$, and $l = 1$ and $m = -1$; and so on.

General angular momentum. $\{|k,j,m\rangle\}$ representation

If $m < j$, we know that exists another subspace $\mathcal{E}(j,m+1)$ made by the eigenvectors of J^2 and J_z with eigenvalues $j(j+1)$ and $(m+1)\hbar$

Moreover, if $m > -j$, we know that exists the subspace $\mathcal{E}(j,m-1)$

Take the $\{|k,j,m\rangle;\ k = 1,2,\dots,g(j,m)\}$ as an orthonormal basis in $\mathcal{E}(j,m)$

Now, we construct an orthonormal basis in $\mathcal{E}(j,m+1)$ by application of J_+

$$J_+|k,j,m\rangle = \hbar\sqrt{j(j+1)-m(m+1)}\,|k,j,m+1\rangle$$

and by application of J_- we obtain a similar, orthonormal, basis for $\mathcal{E}(j,m-1)$

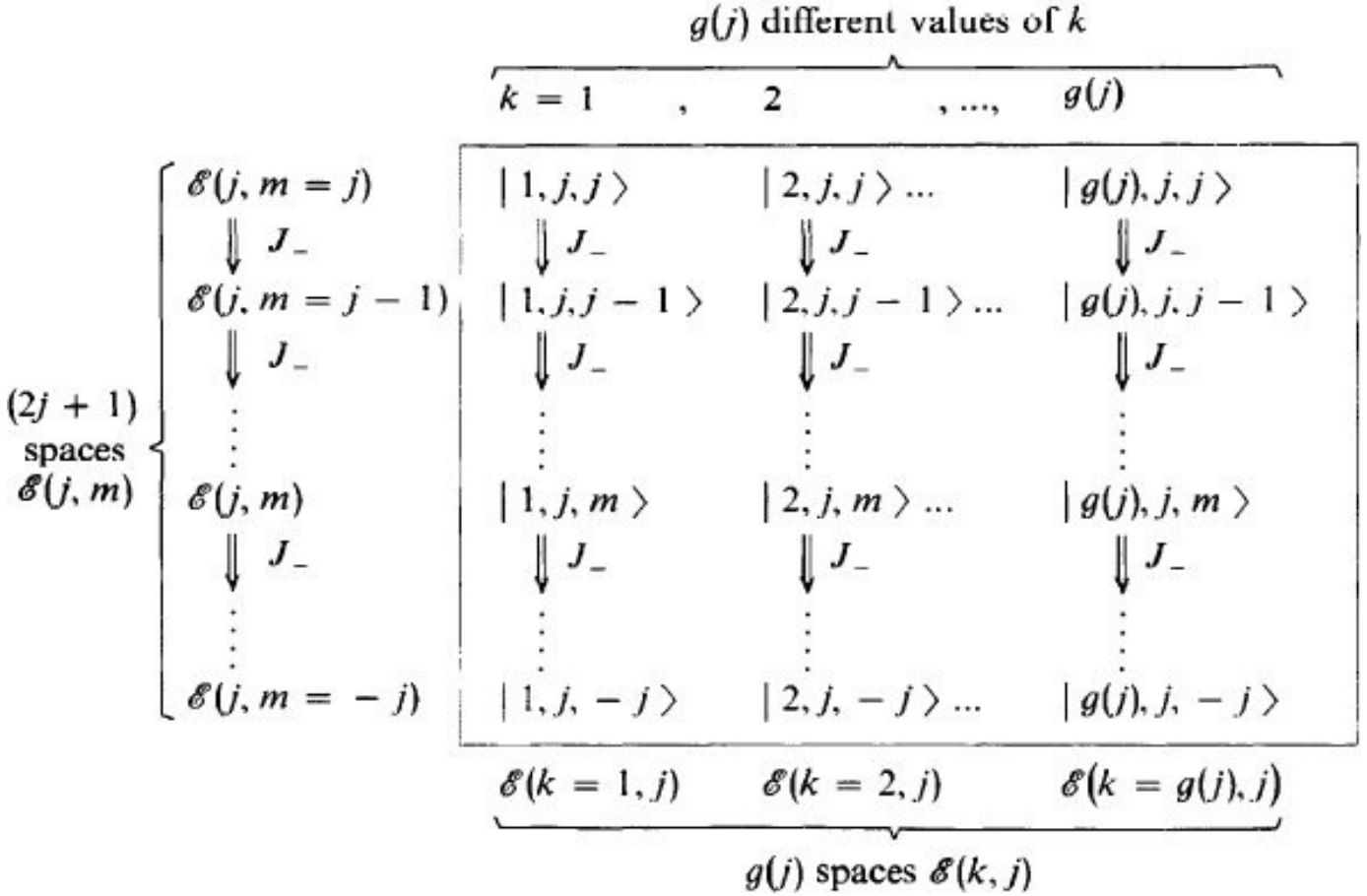
$$J_-|k,j,m\rangle = \hbar\sqrt{j(j+1)-m(m-1)}\,|k,j,m-1\rangle$$

Moreover, $g(j,m) = g(j,m+1) = g(j,m-1) = g(j)$

General angular momentum. $\{|k, j, m\rangle\}$ representation

Schematic representation of the construction of the $(2j + 1)g(j)$ vectors of a "standard basis" associated with a fixed value of j. Starting with each of the $g(j)$ vectors $|k, j, j\rangle$ of the first line, one uses the action of J_- to construct the $(2j + 1)$ vectors of the corresponding column.

Each subspace $\mathcal{E}(j, m)$ is spanned by the $g(j)$ vectors situated in the same row. Each subspace $\mathcal{E}(k, j)$ is spanned by the $(2j + 1)$ vectors of the corresponding column.



General angular momentum. $\{|k,j,m\rangle\}$ representation

- The procedure just sketched has to be applied for each value of j separately. It will generate basis for all the $(2j+1)$ subspaces, $\mathcal{E}(j,m)$, with $-j \leq m \leq j$.
- By construction, we see that $g(j,m) = g(j)$, i.e. the dimension of each $\mathcal{E}(j,m)$ subspace does not depend on m.
- Note that $\langle k,j,m|k',j',m'\rangle = \delta_{kk'}\delta_{jj'}\delta_{mm'}$
- $\sum_j \sum_{m=-j}^{+j} \sum_{k=1}^{g(j)} |k,j,m\rangle\langle k,j,m| = \mathbb{I}$
- In order to fully describe the basis, it is usual to consider additional observable(-s) that commute with J_x , J_y and J_z and moreover form a CSCO with J^2 and J_z .

  Let us suppose that only one observable, A (typically the hamiltonian), is needed to form a CSCO. Now, each subspace $\mathcal{E}(j,m)$ is globally invariant under the action of A.
- The total state space is just the union (direct sum) of the subspaces $\mathcal{E}(j,m)$, where $-j \leq m \leq j$ and j takes on all the values actually found in a given problem. Now, any ket of $\mathcal{E}_r$ can be written as a sum of vectors, each belonging to a specific subspace $\mathcal{E}(j,m)$.

General angular momentum. $\{|k, j, m\rangle\}$ representation

The total state space can also be divided in other ways, for instance considering the subspaces $\mathcal{E}(\mathrm{k}, \mathrm{j})$, in which we fix *k* and *j* (instead of *j* and *m*) and let *m* vary (instead of *k*). Advantages?

- ✓ the dimension of $\mathcal{E}(\mathrm{k}, \mathrm{j})$ is always equal to 2j + 1 irrespective of the value of k and of the physical system.
- ✓ the $\mathcal{E}(\mathrm{k}, \mathrm{j})$ are invariant under the action of $\vec{\mathrm{J}}$ or $\mathrm{F}(\vec{\mathrm{J}})$. (this is so because any of such functions can be written in terms of $\mathrm{J_z}$, $\mathrm{J_+}$, $\mathrm{J_-}$)

This allows to calculate, once and for all, the universal matrices representing an arbitrary component $\mathrm{J_u}$ of the angular momentum (we will see some examples in problems).

For instance in the hydrogen atom, this choice would amount to consider subspaces $\mathcal{E}(\mathrm{k}, \mathrm{j})$, with the same values of *n* and *l* , namely, the orbitals 1*s* (1-dim), 2*s* (1-dim),2*p* (3-dim)....,3*d* (5-dim),..and so on.

General angular momentum. $\{|k, j, m\rangle\}$ representation

Using the subspaces $\mathscr{E}(k, j)$ considerably simplifies the search for the matrix which represents, in a "standard" basis, a component J_u of $\mathbf{J}$ [or an arbitrary function $F(\mathbf{J})$]. The matrix elements between two basis kets belonging to two different subspaces $\mathscr{E}(k, j)$ are zero. The matrix therefore has the following form:

	$\mathscr{E}(k, j)$	$\mathscr{E}(k', j)$	$\mathscr{E}(k', j')$	...
$\mathscr{E}(k, j)$	matrix $(2j + 1) \times (2j + 1)$	0	0	0
$\mathscr{E}(k', j)$	0	matrix $(2j + 1) \times (2j + 1)$	0	0
$\mathscr{E}(k', j')$	0	0	matrix $(2j' + 1) \times (2j' + 1)$	0
$\vdots$	0	0	0	0

$$
\begin{aligned}
J_z \,|\, k, j, m \,\rangle &= m\hbar \,|\, k, j, m \,\rangle \\
J_+ \,|\, k, j, m \,\rangle &= \hbar \sqrt{j(j+1) - m(m+1)} \,|\, k, j, m+1 \,\rangle \\
J_- \,|\, k, j, m \,\rangle &= \hbar \sqrt{j(j+1) - m(m-1)} \,|\, k, j, m-1 \,\rangle
\end{aligned}
$$

General angular momentum. Standard basis

An orthonormal basis $\{|k,j,m\rangle\}$ of the state space, composed of eigenvectors common to J^2 and J_z

$$J^2|k,j,m\rangle = \hbar^2 j(j+1)|k,j,m\rangle$$

$$J_z|k,j,m\rangle = m\hbar|k,j,m\rangle$$

is called a **standard basis** if the action of the operators J_+ and J_- on the basis vectors is given by :

$$J_+|k,j,m\rangle = \hbar\sqrt{j(j+1)-m(m+1)}|k,j,m+1\rangle$$

$$J_-|k,j,m\rangle = \hbar\sqrt{j(j+1)-m(m-1)}|k,j,m-1\rangle$$

General angular momentum.

How can the experimental results about measurements of the angular momentum be interpreted?

The orbital angular momentum l is necessarily integer because $Y_l^m(\theta, 0) = Y_l^m(\theta, 2\pi)$.

Postulating that the electron has an intrinsic angular momentum, conventionally called spin, $\vec{S}$, whose square has as eigenvalue $s(s+1)\hbar^2$ with $s = 1/2$ (it is said that the electron has spin 1/2), that has to be added to its orbital angular momentum to obtain the total angular momentum.

In fact, all the particles carry a spin, with a particular value of *s*, that can be $0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$, which is as characteristic of the particle as its mass or its charge